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H8S COOP

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BITS & PIECES

400K DRIVES

The hottest news is still the 400K drives. I want to take some space to tell you a bit about them. Jan Johnsen, one of my strong supporters contacted me about the 400K drives which use the same controller card, for the H8 and H89. He put me in contact with Dean Gibson of UltiMeth Corporation, who is the author of the software which makes this possible. Let me tell you, Dean really has his act together!

Dean Gibson of UltiMeth Corporation, 24025 Fernlake Dr, Harbor City, CA 90701 (213) 539-4726 developed a new SY.DVD and associated software enabling HDOS 2.0 to be used with these Tandon TM-100-4 drives. These drives are sold as 1 Meg drives! The new driver will handle these and the standard drives in any combo, mix or match, including SY0: which means you can use it as the system drive and boot off of it.

To use this with the H8, the 0-org board must be present (unless you use the D-G board). For the H89, no mods are required. The 0-org for the H8 is needed for side select. You can have up to 3 of these drives for a total of 1.2 megs disk storage! When Heath releases their new double density controller for the H8 and H89, this drive will allow you to get 800K per disc. Presently with this 400K drive and associated software, the same discs are used since 10 hole hard sectoring is used, same as with the standard drives. We have found in the majority of cases the standard discs, if of good quality, can be used. The INIT routine can also call for a MEDIA CHECK, so you'll know if the disc has problems. We have been using Memorex single sided, single density discs, with no problems. I understand Heath's double density controller will use soft sectoring, which will make your old discs obsolete.

So how is 400K possible? What the new driver supports is double side and double track operation with the present controller. This is for HDOS only, since Heath's CP/M BDOS does not support anything but the standard 100K drives. If you are interested in CP/M, contact Magnolia Micro- systems, as their modified Heath CP/M will handle these drives for about 350K on each drive. They also have available a double density controller card for the H89 under CP/M alone. It would seem if someone like Dean would write a new driver for this card, HDOS could also take advantage of the 800K/disk, or 200K for the standard drives. Anyway, if you are interested, contact Magnolia at (800) 426-2841. I understand some individuals are writing a mod for Heath's CP/M's BIOS and BDOS to allow these drives to function under the Heath system as well, to take advantage of the 350K for CP/M. I also have been told that the new release of Heath CP/M will take care of this drive, since it is the same drive they will be using. In any event, this drive will

function one way or the other under CP/M now (by going Magnolia) or in the future.

These drives step at 3ms, and SET options are available for each drive as to # sides and tracks, along with system SET options such as motor idle time. The new SY.DVD which replaces the Heath SY.DVD offers the following features:

A 35% reduction in time to load large programs and in copying large files using PIP.
Individually SETable step times for SY0:, SY1:, and SY2:.

The ability to SET the time intervals after a disk I/O operation that the head stays loaded and the motor stays on. This allows the head to remain loaded between rapid I/O operations, reducing head and media wear.

Recording of the step times in the boot track, resulting in up to a 50% reduction in boot time for fast drives.

Ability to preform a media check during INIT, eliminating the need for TEST17, except for drive check-out.

Support for 80 track per side drives, including the ability to automatically detect and read, including boot (but not write) a 40 track diskette in a 80 track drive.

Support for dual-sided drives, including the ability to automatically detect and read, boot, and write a single sided diskette in a double sided drive.

Improved error recovery which temporarily increases the step time during the error retry of a disk operation, allowing the step time to be SET to give the fastest usable seek rate, and still handle an occasional error.

It may be worth noting that this driver can be used even without the TANDON drives, with the standard drives, to take advantage of some of the features such as individually setable step times, and the first few points mentioned under features.

You will need at least one standard drive to configure and set this up. All standard HDOS functions still apply, and PIP is used to do all disc copying.

Hats off to Dean Gibson, of UltiMeth for this long awaited brainstorm. I have been using mine for over a month now, and it's fantastic. Dean will be coming up with more stuff in the future, see NEW STUFF for another item by Dean.

Anyway, I have already sold over 30 of these, and will still be taking orders for the drive with software for \$495 for another month, and then the price will go up to \$550 or so.

It may be well to note, this drive normally sells for about \$600, and that is what Heath is currently planning to sell them for. It is also worth noting, there are other drives on the market with this capability, such as MPI, but all my investigation (and apparently Heath's too) tells me to stay away from them.

One point I want to pass along to H89 owners, this drive if used as SY1: with the H89 does not have to be turned on when booting the system like the standard Heath drives. For some reason, booting with the drive turned off will not destroy the SY0: disc, which is the case now with the present drives. But it is still not a good practice to send signals into a chip without having power applied to it as well.

Also to pass along for 2 and 3 drive systems, if you are having problems with seek errors, it is important to have the terminating resistor pack installed on the last drive only, not on all 2 or 3, as too much loading occurs, which will result in problems.

Just as I was ready to go to press I talked to Dean Gibson and he told me UltiMeth Corporation, in co-operation with Magnolia will be offering an HDOS driver to work on the Magnolia double density controller. Magnolia's controller will handle up to 4 5" drives and 4 8" drives, plus the disk controller card presently in the system. This Magnolia double density card for H89's will also allow any standard 8" type drive to be used with the H89. They are aiming at a September 1st delivery date on this software, and who will be distributing it is not certain at this point, I'll keep you up to date on it.

UPDATES ON WORD PROCESSOR ARTICLE

Dave Gomberg of Lum Buckles, 410 Townsend St, San Francisco, CA 94107 wrote to give me his phone number and address of his work place where you are more likely to catch him. His phone number is (415) 495-0339. Dave handles Autoscribe, and has available such things as Client Billing and a Mail merge to run with Autoscribe, to make this wordprocessor more versatile and complete. Assuming you have MBASIC and HDOS, ASPAK to allow access to AutoScribe will cost \$100, MailMerge goes for \$100, and Client Billing goes for \$200. Contact Dave if you are interested in more details.

I was given some information on TEXTIT. I was told that it was updated for HDOS 2.0 and some new features added. It cost \$69.95 and is sold by Great Western Software, 1533 W. Mosurri #4, Pheonix, AZ 85015. The phone (602) 265-8315.

ON DUALPORT

I now have information on how to obtain DUALPORT as mentioned in last months H8SCOOP. Rick Kerble is the author, Sunflower Software, 6304 W. 76th St, Prarie Village, KS 66204. Rick tells me many local HEATH stores handle this, and he suggests you get it from them if you are interested.

PRAISE ON Z80 BOARD FROM TINIVUS

See MAY H8SCOOP page 7 for info about this H8 Z80 mod board. Tor Tverre writes in "I have received my HA8-8 board. I have also installed the Z80 board from Tinivus in Germany. This combo works perfect. Any type of software; Mychess, PIE Z-89, and what have you, works perfect. I would like to recommend Tinivus's Z80 board. It is beautifully made, just like a German Mercedes car. He has used gold connectors where possible, and the board is shining like a jewel. It is almost too beautiful to hide away in a computer. To install it is done in 10

minutes and no soldering or anything is required. And best of all, the board is not only assembled, but burned in for 23 hours. I just mention this to give an impression of the care that he has put into the production of this board. I feel that this is the cheapest way to go Z80, and you can have it NOW, not some time this fall."

APPLICATIONS FOR DON LONG'S PC-12

I have been asked about converting some of my GOODIES project articles from the H8-2 parallel board to the PC-12 parallel board. Don Long has already converted the Selectric article to his board, and may eventually convert some of my other projects, which should not be too hard, since only the initialization will change.

If anyone has converted any of these over already, drop me a line and let me know. If and when I get some time I will also work on this and pass along the info. If anyone has done any other neat applications, let me know so we can share it with all.

STOP THIEF!

Jim Gillogly, 2520 Chard Ave, Topanga, CA 90290 tells me his H89 has been stolen. In my opinion, this is the work of only the lowest SCUM of the earth, and I hope they are caught. Any information leading to the arrest and conviction will result in a free lifetime subscription to H8SCOOP, plus a mighty good feeling. If anyone is trying to push this stuff off, and anyone becomes aware, please take appropriate action. This kind of SCUM is not needed in the world, and the sooner they are behind bars, the better! On June 10, 1981 two computers were stolen from Jim Gillogly's home in Topanga, California (Jim's the guy who writes that good software distributed by Software Toolworks).

Heathkit H89 Serial #00908 with 64K RAM, RS232, FBE centronics interface card, Outboard 5.25" drive, plain black case. Identifying marks: Gummed graphics symbols on the front of keys, braided cables from the power supply (!), grey paint on the front of the keyboard worn off, below the return key.

ATARI 800, Serial #081-1577, 32K Microtek RAM cartridge, Editor/Assembler, BASIC, Star Raiders modules, 13" RCA color TV, Serial #048311463.

If any of these are spotted (preferably with Jim's Kenwood TS-520 radio nearby), please contact the L.A. Sheriff's office (213) 456-6652 and tell the detectives about file #581-03337-1035-507.

SUPPORT FOR HEATH'S PASCAL

It appears Softech is now offering better support of the PASCAL Heath is selling. Ralph Munroe called to pass along this info to me. Anyone having the Pascal system 1 discs having problems such as with the H14 printer as described in May 1981 H8SCOOP can get the fix free. Contact John Tennant of Softech and get a return goods number. Then send that number with the faulty disc software and will be replaced. You can call (714) 578-6105 or write at 9494 Black Mountain Rd, San Diego, CA 92126.

NOTE ON H47 DRIVES AND HEATH CP/M

Walt Bilofsky writes "H47 owners running Heath CP/M may have noticed that it takes a long time for the system to warm boot following a ctrl-C or the end of a program. The H47 master drive contains jumpers which set the default disk density, and changing these can reduce the wait. Jumpering YJ and FG reduces the ctrl-C response time from 9 to 5 seconds. But beware: HDOS will not boot under this setting! (I don't understand why not, since the effect of those jumpers are supposed to be completely internal to the drive, but it doesn't work.) If you're running HDOS, the best you can do is jumper YJ and ZF; this keeps CP/M slow but reduces the HDOS boot time from 23 to 17 seconds.

MISCELLANEOUS

Don Woodruff wants to know (and so do lot of others) if there is a way to modify HDOS to allow more than 255 discs to be recognized as many of us are presently housing libraries of over 400! There was definitely some short sightedness in making 255 a maximum number. Any feedback, direct it to H8SCOOP.

Don also says there have been numerous letters regarding people forgetting to either turn on or turn off all equipment at one time with consequent problems. The solution is to use a multi-outlet box and turn on and off ALL equipment at one time. This is an excellent idea. I have been using the Radio Shack TRS-80 (who said that!) line filter which combines this feature with line filtering. I have also added 3 led's to my H17, one for 5V supply, one for 12V and one for RUN. This way I can instantly see if it is on, and when the motors are running even if they are not selected, plus if I have problems, I can instantly narrow it down to power supply or something else.

Burt Hulland says he has been using PRUF-REEDER from IST Software for \$39.95 to proof read his text. He says it's a little slow, but for \$39.95 it is a pretty good buy. I wish I knew more on that one, but that's all he sent me.

From John Croft; says he will readily endorse TOUCH TYPING TUTOR from Xtra-Soft (see June H8SCOOP). He says the tutor is good, entertaining, and he has picked up speed by staying with it. John also sends a thought for any like him who use tri-focal glasses. He had his optician make a special pair of glasses for use when seated at the CRT. "Sure beats nodding the head up and down like a dunking bird". John also says the documentation provided in the June Microcomputing on DATALOG by Ray Massa of Keyboard Studio is mostly the same that accompanes the advertised 5 1/4" disc for his \$10 price, but 14 pages of entries done by hand verses a working product makes it all worthwhile to buy the disc, and it is a good program.

NEW STUFF

NEW DISK CONTROLLER FOR H89

From CDR Systems, Inc, 7667 Vickers St, Suite C, San Diego, CA 92111 comes a new controller board for Heath H88/H89 micro's to double the disk storage on Shugart compatible disk drives. The same drives

and discs can be used. The board (FDC-880H) runs under CP/M 2.2 and can handle up to 4 Shugart compatible type drives, 5 1/4 or 8" simultaneously. It also handles single and double sided operation, single and double density encoding. It converts the hard sector discs to standard soft sector double density, and will install in any I/O slot in the micro without any mods.

Price is \$695 and includes 10' interconnect cable, zero origin conversion PROM, boot PROM with floppy diagnostics and manuals. Delivery 30 days. For more info, write or call Marc Brooks at (714) 275-1272.

COMMUNITY COMPUTERISTS DIRECTORY

The Community Computerist's Directory, POB 405, Forestville, CA 95436 (707) 887-1857 has a new directory dedicated to helping computerists and non computerists find each other across the boundaries of computer jargon and machine oriented groups and publications. It can be used in finding people with similar interests or projects information or expertise, or someone with a computer they are willing to share etc.

It is similar to a telephone book with these differences: It focuses on the driving force behind the info reveloution, the people and organizations who are producing, buying, using and experimenting with computers; and includes a statement written by each participant describing their projects, interests, needs and resources. Adds can be placed by anyone in the white page or yellow page section at a cost of \$10 to \$40. For more info, contact CCD.

NAUTILUS SYSTEMS CROSS COMPILER FOR FORTH

Nautilus Systems, POB 1098, Santa Cruz, CA 95061 (408) 475-7461 has a high-quality, low-cost cross compiler for the FORTH language that will run on an H89. They write "Our cross-compiler provides the most convenient method to implement FORTH on a target computer, or to extend/modify FORTH on a host computer. This cross-compiler has many other unique features, including: automatic forward referencing to any word or label; capability of producing headerless code; capability of producing ROMable code with initialized variables; a load map; and a comprehensive list of undefined symbols. 32K is the required minimum memory size. The price is \$150, call or write for more info.

MORE FROM THE GENIUS OF ULTIMETH

More from UltiMeth Corporation, 24025 Fernlake Dr, Harbor City, CA 90710 (213) 539-4276. MTRHEX and MTROCT are replacement monitor ROMs for the H89, providing increased HDOS processing speed and additional debugging capabilities. The following features are provided:

The monitor clock interrupt processing routine has been rewritten, giving an effective 16% CPU speed increase to HDOS programs.

The clock value in memory (TICCNT) has been extended to a four byte counter.

Additional monitor debugging commands have been provided to display and alter the registers, to read and write to I/O ports, to single step programs, and to restart HDOS.

Full compatibility with HDOS and CP/M has been maintained. Each ROM includes all of the common HEATH H89 entry points (except for cassette tape processing). The monitor is available in either an octal (MTROCT) or hexadecimal (MTRHEX) version, which you must specify when ordering. The ROM is available for \$50, or \$53 for California orders which includes shipping and documentation. Source code is NOT provided.

Refer to BITS & PIECES for info on the 400K SY.DVD driver written by Dean Gibson of UltiMeth Corp., an excellent piece of work. Dean has already established his quality, we have no fear to buy what Dean has to offer.

UPDATE FROM HULLAND ENGINEERING

Notes from HULLAND ENGINEERING, 555 Broadhollow Rd, Melville, NY 11747 (516) 420-0970. Burt Hulland is the author of the HDOS 0.1 operating system as mentioned in last months H8SCOOP (as 0.6, incorrectly). He notes "if you have an H8 with 64K of RAM, you will have noted a problem, --the same one HDOS 1.6 has with that same configuration. I am now shipping release BH001.02 which fixes that. Anyone with the .01 release can get a free update by returning the original disk, or if they prefer, a patch is available for the original version."

Burt also tells me the CTRL R function of PIE or PIE8 does not work with HDOS 0.1. He has a patch available which will allow CTRL N to be used for this function, if anyone uses it. Write Burt for this patch, and specify which version and release of PIE you have.

Burt also has a device driver for the MX-80 that allows all functions to be accessed using only ASCII printing characters (no control- or escape- codes) so that it can be used with any editor. It is available for \$15 from HULLAND ENGINEERING and is supplied in two versions, one for the H8 with H8-4 or similar interface (including H89) and the other for the H8 with the old H8-5 interface.

Burt has a new addition which will run under HDOS 0.1 ONLY, which allows any device named LP: to print everything that appears on the console terminal. It is turned on and off by CTRL R. Sounds similar to the CTRL P function under CP/M. Supplied as a .DVD file, it sells for \$15, or get the MX-80 driver and this .DVD driver on one disk for \$20. There will be more coming from Burt, stay tuned to H8SCOOP for the SCOOP.

MORE ADVENTURE

From CALIFORNIA DATA BASE, 4554 Cristy Way, Castro Valley, CA 94546 comes several adventure type games in MBASIC requiring 48K HDOS. This package sells for \$29.95, with a 20% discount for readers of H8SCOOP (mention you saw the info here).

THE CAVERN OF DOOM places the player at the entrance to an underground cavern. Great treasure awaits the thoughtful adventurer, while certain death is waiting for the foolhardy. LEYGREF'S CASTLE is an ancient ominous castle wherein lurk numerous monsters, vast treasures and incredible magic. THE TREASURE OF THE SIERRA MADRE is a western adventure which rewards the creative traveler.

THE DREADED DUNGEON OF DESPAIR contains well over 100 different named monsters, poison gas and hideous demons. Magic and mystery run rampant as you explore this dungeon which is apparently bottomless. Many brave adventurers have entered the dungeon but none have survived.

LOAD-N-GO

From Skip Barron of Mako Data Products, 7441-#B N. Red Gum, Anaheim, CA 92806 is HDOS 2.0 compatible LOAD-N-GO. This version contains some new features and improvements from the past 1.6 version. LOAD-N-GO 2.0 is a Turnkey modification package for HDOS 2.0 issue 50.06.00, providing for auto boot-up, auto-mount of validly installed discs (no hang-up), and do AUTO.CMD or RUN AUTO.??? on both cold AND warm boot. As an extra bonus, HDOS 2.0 "Doctor-Kit" will be thrown in, a series of 7 programs which perform the most popular "patches" to the system, with full error checking. The price is \$16.95 ppd, USA, Foreign orders add \$2.00.

INFO ABOUT NEW HEATH STUFF

Scheduled for the winter of this year is the firmware revision for the H19 which will add the missing diagonal graphics character, and also allow the H19 to operate reliably at 19.2 baud.

They are also planning release of the Winchester 10 meg hard disk, with an 8" floppy in one cabinet for the H8 and H89. It will be sold in this configuration because you will need something for backup. Sold as the H67, expect a price af about \$6,000. Just as a point of interest, I understand they will no longer be selling the dual 8" drive because of lack of sales. The double density board for the H8 and H89 should also be released, costing about \$350. They will be reportedly selling the drive, the same one I offer for \$465 for \$600. This will require new soft sector 5 1/4" discs, and will allow this Tandon TM-100-4 drive to store 800K. It will be sold as the H-37.

Of course with these new drives, we can expect HDOS 3.0 to be released to support them, and a new CP/M. HDOS 3.0 will also be a 0-org HDOS, just like CP/M, if they can get it in on schedule. This will allow full use of your 64K memory.

The Z80-CPU should be on schedule for latter this summer. Their new H-25 high speed dot matrix line printer should also be available for the summer for about \$1000, printing at 100cps. It is also reported that Heath will market the DC Hayes smart modem which is an auto-dial/auto answer/originate/answer 300 baud modem, for under \$300.

ATARI 800 MEMORY EXPANSION

For those of you who have the Atari 800 (the same model stolen from Jim Gillogy) for \$79 you can get the 16K memory expansion from Lisatronics, 17955 Skypark Cir, Irvine, CA 92714. Phone (714) 557-6536.

SECURED COMPUTER SYSTEMS OFFERS PRODUCTS

Secured Computer Systems, 12011 Aclare St, Cerritos, CA 90701 (213) 924-6741 offers a 16K RAM expansion card for the H88/H89 type computers for \$100. This is functionally the same as the Heath board for \$150. If you also need the PROM upgrade it will

cost \$50 when ordering with the 16K RAM expansion card.

They also offer the Anadex DP 9501 line printer which also contains a 2K buffer for \$1475, and are handling an 8" disk system using the Remex disk drives for \$2795 without the disk interface card. They will be offering several new products this summer for the H89. Contact them for the full SCOOP.

SOFTWARE TOOLWORKS UPDATE

Walt Bilofsky of The Software Toolworks has three new software products available. MUNCHKIN is an arcade-style action game selling for \$19.95. You run through a maze evading creatures which try to gobble you up. Turn the tables by reaching a force point, and you're energized to catch and destroy the baddies.

SPACE PIRATES for \$19.95 is also a video game. You've drawn the duty at the Delta Pegasi IV space station, where grain barges await transport to the starving hordes of Earth. Raiders swarm in to steal the barges, and it's up to you to fight them off with your remotely piloted laser equipped robot ships.

LEARN TO PROGRAM IN BASIC for \$29.95 on the H89 or H8/H19. This course turns your computer into a computerized classroom for this course in BASIC programming for beginners. Uses BH BASIC, but the material covered applies to MBASIC as well.

For more details on this, write Walt, visit your local Heath store, or see the latest BUSS directory for Walt's complete catalog.

SOFTWARE FROM GWS

From Great Western Software, 1533 W. Missouri #4, Phoenix, AZ 85015 (602) 265-8315 comes a variety of software. TEXTIT is a menu driven word processor under HDOS which sells for \$69.95. It features Help menus, underline and bold print, macro definition/expansion facility for form letters and more.

MUDos for \$99.95 is a Multi-user MODEM package which turns the H89 into a powerful multi-user dial-up computer system. It supports the BC103 auto-answer modem. Any number of users can be registered (each has a unique USER-ID and PASSWORD), although only one online user is allowed at a time. Each user is assigned a security level and home directory. Many more features, this one looks pretty good.

Q3, Interactive Modem Terminal Link is a powerful modem communication tool with features like auto answer and dial when used with the BC103 modem, built in help function, incoming non-printing characters are shown in reverse video and much more.

INV1 for \$59.95 is a high speed multi purpose inventory system. 10 data files which can be changed in a configuration menu can store up to 2000 records on a single drive H89 system, with record access time of less than 10 seconds. The printout indicates Reorder list, Stock check list, Master inventory listing, Stock room shelf labels, and Sales floor price labels. The online system automatically adjusts inventory levels from

invoice. All prices, descriptions, etc, come from data base.

For \$29.95 INV2, a display orientated information storage system with built in HELP function. Up to 10 user definable data fields can be created and searched or Totaled, according to exact matches in a particular field, substring matches in any field, less or greater than in any field.

VEDIT for \$19.95 is a video editor for fig-FORTH featuring line insert or delete, line pick and put, character insert and delete, scrolling forward or back by screens.

BIBLIOGRAPHY UPDATE

The May 15th issue of The National Software Register, formerly The Software Trader, reported that the June 1981 Radio Electronics magazine had an article on a simple, no calibration \$60 answer/originate modem. I looked it over, and it looks quite simple, and very economical.

Again, Microcomputing pulls through. The July 1981 issue had a book review on page 216 on the book "H-8 PROGRAMMING FOR BEGINNERS". Page 152 carried an in-depth article on WORDSTAR, a word processor under CP/M, for those interested. Page 92 has a short article by D. C. Shoemaker on "TRS-80 Graphics on the Heath H19", for those interested in converting TRS-80 programs to the H19. Page 66 carried an article by Tom Jorgenson, "Dissecting the HDOS Diskette", an article about some inside info on HDOS. And page 80 carried an article on adding the H77 to the H89, "Shift into Extra Drive On Your Heath".

OFFERS

I make all software available on an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy discs. You take your chances. At my prices, you really can't go wrong. I try to get software out as fast as possible so you can get the newest stuff fast, and as thus, don't get a chance to carefully screen and test everything. To do so, would delay all software at least 3 to 4 months. I do the best I can and sell without assuming responsibility on the accuracy or runability of a program or disc. I don't think I've let anyone down in the past, and will continue to do my best not to.

The following are past offers made by H8SCOOP, which are and will continue to be made available to those interested readers. Note I have dropped a few items from the list, and as time goes on, more will be dropped, so my inventory doesn't go crazy around here.

9/80 FILE DUMPS & UTILITIES	\$10
11/80 PICTURES, LABELS, WORD PROCESSOR	\$15
QUIKMAIL MAILING LIST PROGRAM	\$15
"C" SOFTWARE STUFF DISC	\$10
12/80 PICTURES & UTILITIES	\$10
1/81 H19 GRAPHICS PROGRAMS DISC	\$10
2/81 PLANT, SWARMS, STARWARS, UTILITIES	\$10
3/81 POTPOURRI SET(2) MISC CASSETTE CONV	\$15
3/81 ADVENTURE SET(2) PIRATES, LOST DUTCH.	\$15

W H O ' S W H O

4/81 THINGS--H19 GRAPHICS, BIORYTHM, UTILITY \$10
5/81 HDOS 2.0 PATCHES \$10

ON MX-80 PRINTERS

I am no longer handling the CP/M NPS COBOL package. K. Stephen Tinius, HQ USEUCOM Box 741, APO NY 09128 has a new fixed version which runs on org-0 CP/M. He says he can get the source files as well, but they are written in INTEL's PL/M-80 and need the PL/M-80 compiler and ISIS-II operating system to recompile or adapt to Lifeboat's CP/M. He will give it to anyone interested, just supply 2 blank disks, and return postage. He also has the SARGON chess game for HDOS, including source files. It runs on the Z80 only, but the source has equates for the Z80 opcodes, and it will assemble with the standard Heath ASM.ABS. It's from Peter van den Hamer of the Dutch HUG. Evert Jan Stokking, also from Holland, sent Stephen a superb graphic Reversi (Othello) written in Tiny Pascal, source included. If interested, contact Stephen Tinius on that one as well.

This month there will be several disc offers to make up for last month when I did not offer any. For \$10 the 747/GRAPHICS.MB disc is a 747 flight simulation program taken from August 1980 INTERFACE AGE magazine. Both BH BASIC and MBASIC versions are supplied on disc. This disc also contains some excellent graphics games, Battleship, Jump, and Shoot. I'm sure you are all familiar with Battleship, this version the computer hides the ships and you must blast them out of the water. Jump is another excellent and challenging graphics game where you jump over pegs in a board to eliminate all pegs except the center--not as easy as it sounds. Shoot is a game of firing a mortar shell at a target under different gravitational fields. An editor for doing letters or what have you, a page at a time, is supplied also, which runs under MBASIC. This entire disc is graphics and MBASIC is needed.

The next disc, MISC I Disc has some challenging games, if you want to call them that, which have appeared in magazines. OREGON Trail and ATOM20, along with a Home Budget program. This one will sell for \$7.?

And lastly, ODDS & ENDS is a collection of programs. There is a super calculator program which requires 56K (all these are also MBASIC) and will do almost anything and everything, including metric conversions. There is also a home finance program which figures installment loans, bank interests, and ballances the checkbook. A File program to keep track of a Ham Radio Club, but could be modified to fit your needs. A SORT program to sort ASCII files that will not fit in memory at one time, and to go with it a MERGE program to combine two files into one. Also a very excellent random file generator/maintenance program. This will create a file for you, which you specify field descriptions and lengths. You can then add, change, delete, list, display the file. All the random I/O is automatic, you just specify parameters. This disc will go for \$10.

Since the MX-80 is a very popular printer, due to its low cost and versatility, I will devote some space here so those going to the MX-80 will know who to contact for problems and printer configurations.

This printer has been successfully been connected to the H89, and H8 via H8-4 card serial, and also the H8-2 card for parallel operation, take your pick. Douglas Arndt, 1111 5th Ave, N.E., Watertown, SD 57201 has configured his MX-80 to run parallel with the H8-2 card using an H8-5 device driver.

Jim Czebiniak, 199 Vly Road Ext, Schenectady, NY 12309 (518) 869-8723 has been running his MX-80 on the H8-4 serial card, with help from H8SCOOP info received from Jeff Sohn, who is running his on the H89. Jeff Sohn, POB 118, Sheboygan Falls, WI 53085. Jeff used the H14 device driver, and noted that in his case (as was the same with Jim) that pin 3 of the printer is tied to pin 3 of the computer (RS-232 received data), pin 7 to pin 7 (signal ground), and pin 20 printer to pin 4 of the RS-232. It must be pointed out that in some cases, pin 20 must be tied to pin 20. I don't know why the difference exists, but make note of it.

The printer should be configured to 9600 baud, and the following switch settings have been used and seem to work best for printer configuration. On Dip #1 in the printer, SW# 1,2,5,6,8 should be set to ON. Dip #2 in printer should have SW# 1,2 ON. On the serial interface board in the printer, SW# 1 should be ON. All others should be in the OFF position. The printer port is usually 340 for the LP.

I also understand there are various device drivers available for this printer, but at this time do not know any more about it. Oh Yes, Set printer interrupt to level 5.

James Czebiniak is still having two problems with his MX-80 someone may be able to help with. It will not set 8LPI and will not print graphics characters. He says "When I print CHR\$(151) (which is a graphics character) it appears that the high order bit is dropped and I get a standard ASCII character".

REQUESTS

Tor Tverre, Monradsgate 3 A, Oslo 5, NORWAY, wants to buy a new printer for letter quality and is considering the Paper Tiger 460G. He wants to know if this printer or any others will either support the Norwegian alphabet which has 29 letters in the alphabet, or if any printer is available with a custom ROM to obtain the extra characters needed for the Norwegian alphabet. If anyone has any info on this, contact our foreign friend, Tor.

What he is mainly looking for is a dot matrix printer which contains a separate ROM for the character generator function. This way only the

ROM would have to be changed to change the character set.

I have had some requests for H17 controller boards. If you have an H89 and purchased the H17 (which is a better deal than the H77) and you want to get rid of some H17 controller cards, kit or assembled, contact H8SCOOP with information.

A request from D. C. Shoemaker, 2000A Foxridge, Blacksburg, VA 24060. D. C. wants to know if anybody has given any thought to some sort of conversion to an RS-232 printer port from the H88/H89 cassette interface board. Since most of the H89ers have that board sitting around gathering dust, if someone has, or can come up with a mod to make a RS-232 port, it would be a very valuable piece of info to make available.

Frank Rinkleff, 2036 E. Ridgeway Rd, Waterloo, IA 50702 has a C. ITOH Starwriter printer and needs a device driver to make it work. If anyone has this printer running, write to Frank and let him know what you are using. I just came across a little paragraph in May 1981 U-HUG newsletter that Russ Scholz has some documentation available on the ITOH STARWRITER printer. It describes how to make an interface cable, how to set the switches, and how to patch the LPH24 device driver to get it to run nicely on HDOS. This may be your solution, Frank, and to others. Perhaps John Merrill, 3249 Lincoln, Ogden, Utah 84401 621-4492 or 777-3981 (no area code given) can help you locate this info.

Gregory Louie, 1430 Rock Haven St, Monterey Pk, CA 91754 (213) 576-7744 wants to know if any work is being done on an 8" controller for the H8 that is CP/M compatible, with the real origin 0. This would require a real Floppy Disk Controller with the BIOS to drive. If you have any info on this, contact Gregory.

Bruce Bennett, 299 Smith Hill Rd, Utica, NY 13502 wants to know if anyone has info on making the IDS 460 printer handshake properly with the H8 under CP/M. His computer/printer system works well at 300 baud, but as 1200 baud or faster rates he loses information--apparently because the H8 doesn't understand the signal given by the printer to indicate that the buffer is full.

Charles Bracher, 6221 Fenley Dr, Huntington Beach, CA 92647 (714) 847-5443 has had problems with his 16K Heath static memory board and had a major job in trying to locate the chips for it. He wants to know (and so do lot of others) if anyone knows of a source of these static RAM's.

TECHNICAL FORUM

ON DISCS AND DRIVES

Back to the issue of double sided disc use from single sided discs. It appears that the major objection is only the flipping of the discs which causes rotation to change. Other than that, such as for double sided drives like the one I mentioned in JUNE H8SCOOP, fine, since the direction remains the same. And it seems most discs are good on both

sides, even though sold as single sided. Investigation confirms that when a disc is sold as single sided it is tested and certified for single sided operation, but the opposite side may indeed be good and valid for data retention.

"I would only wish to point out that the pressure pad wears on the side that's normally up and not used, and to flip the disk means that that pad, with all the ingrained dust and iron oxide particles, will then be abraiding your normal data surface. Over months of use, the results are self-evident. For the \$2.25 or so you save, the risk is hardly worth it."--D. C. Shoemaker.

On the other hand, which I question from looking at my few discs I have experimented with, comes a comment from Don Woodruff: "A MEMOREX employee tells me that the liner in the disk holder should prevent hair, dust, etc. from fouling the head regardless of what direction the disk spins. And, with a normal maintenance program of cleaning the head with a swab and alcohol, no loss of data or abnormal head wear has been experienced."

Now on CLEANING, as was pointed out a few issues back, DON'T use the disc cleaners! From D.C. again, "In a word, DON'T! After three years of hard use, I have yet to see the first trace of dirt on the heads. I have twice used a cotton swab moistened with denatured grain alcohol (NOT rubbing alcohol, which contains oils) and got not a trace of dirt. The risks of head damage far outweigh the potential benefits. This assumes that one keeps the discs themselves clean. Some folks may not have noticed that the read head is positioned on the bottom of the disk, with the pressure pad on top. This means that it is the bottom surface that must be kept clean. Since a spinning disk generates a static electric charge, you have to be careful never to lay the disk on anything with the label up, since the disk surface will attract any dust, and the head will then grind it into the surface."

On the subject of DISK HUB REINFORCERS, suffice it to say that feedback indicates that they are not worth the money. The disc surface will eventually wear out anyway, and this usually happens well before the center gives any problems. I have hundreds of discs here, many don't get used much, but some have been used heavily since the drives were introduced by Heath, and have only had one problem, and that was my carelessness. So the bottom line is, why spend the money if you don't have to?

I've been told that on the 400K Tandon drives, which are capable of 1 meg storage, it may be worth using the disc hub reinforcers for the discs used in this drive. When working with double track, double side, and double density (to come) things become critical, these may well be worth protecting. Again, it's strictly up to the user.

On the RESET function, Burton Hulland, the author of HDOS 0.1 points out "I have examined the drives and find that there is NO effect on the head or the pressure pad. The head never retracts (whether the drive is selected or not) and the pressure pad for the head is automatically raised when the door is opened. What some users have seen is a foam

pad that holds down the disk envelope when the drive is selected, and which does not retract when the door is opened. While there is some extra wear on that pad when a disk is inserted with the drive selected, it is very minor. . . The bottom line is: Don't worry about using the "M" command to change disks. One warning however: The disk should NOT be left in with the door closed after an "M". Since the disk is rotating and the head is loaded under these circumstances, head and disk wear will result." The "M" command is under the Hulland Operating System, which can be thought of as the "R" command under HDOS for this discussion.

On DRIVES, Walter Scott tells me he has had an intermittent problem with a drive. The trouble could not be located and caused a large number of soft and occasional hard errors. Lubricating his drive quieted it considerably, but didn't solve the problem. Upon calling Heath about their policy of drive repair, he was told they have an "exchange" policy where they exchange the defective drive for \$236.50! Keep in mind that a new drive can be purchased from FDS for \$235 and you still have the old one for parts!!

Walt called Siemens Corp in Anaheim, CA and was told they will repair it for a flat fee of \$90, which covers all parts, labor and a 90 day warranty. The turn around time is about 2 weeks, and it is shipped back COD. I wanted to print that in hope of saving others money--Maybe you can use the savings to buy hub reinforcers!

I think this subject has been beaten to death, and will leave it at this. I just wanted to present as much info as possible so you can make an intelligent decision.

GOOD GUYS and BAD GUYS

GOOD GUYS--D. C. SHOEMAKER, 2000 A Foxridge, Blacksburg, VA 24060. I have received plenty of positive feedback about the Major in the past and felt he deserves mention. D. C. has been a major supporter of H8SCOOP from the start, and has had numerous articles appearing in Kilobaud and Byte. I received many notes from fellow H8 and H89ers telling me how D. C. has helped them. Tor Tverre writes "Wherever I write I find out that he has already been there before. He is putting a tremendous amount of time and effort into helping people all over the world getting the most out of their H8. I feel this (mention in GOOD GUYS COLUMN) could be a good way of letting him know that we appreciate what he is doing for us." I'll buy that. I want to take this space in personally thanking D. C. for all he has done for all of us, especially for H8SCOOP. Hats off to Major Shoemaker.

BAD GUYS--MARK GORDON COMPUTERS, INC. It turns out that Earl Albin who recommended Mark Gordon Computers (MAY H8SCOOP pg 5) now is saying forget them. Another letter from MSgt Lawrence Seymore says about the same thing. It appears they are not delivering on schedule, and ignoring communications to resolve problems.

Lawrence tells me he ordered an MX-80 in January and never received the printer, nor had any positive results in getting satisfaction. He finally wrote them "Well, it has been four months since I placed this order and I still have not received the printer or interface. In addition to that sad fact, I have received only one piece of correspondence from you, ever, since I placed this order: an acknowledgment of the order only, which I received at the end of January..."

Earl writes that when trying to use their toll free number to locate his problem, they would not talk to him unless he called the regular long distance number.

This kind of place should be forever avoided. I can see where problems may crop up, but to cut off communications is without excuse. Patronize companies and individuals that care, Pacific Exchange for discs is one of them, and stay away from joints like MARK GORDON COMPUTERS!

CLASSIFIEDS

FOR SALE--H14 complete with manuals and spare ribbons. Good working condition. \$400 and I will pay shipping in 48 states. John Bennin, 620 Hi St, Baraboo, WI 53913. (608) 356-8400 or 356-7202.

FOR SALE--Two 8K memory boards, \$65 each or both for \$120 (Postpaid USA). W. Craig, POB 311 Grayson, KY 41143.

FOR SALE--H8-5 Serial/Cassette board with recorder, cables, operating system software, and all documentation, \$100. Call Chip Louie, (213) 701-1044 days or (213) 576-7744 evening.

WANTED--H9 video terminal in good working condition. With connector cable and manual. Either modified for upper/lower case conversion or standard keyboard. John Croft, 28016 Green Oaks Dr, Eugene, OR 97402. Phone (503) 688-5877 8AM to 10PM PDT.

FOR SALE--Four 8K Heath memory boards, two or more for \$50 each, or \$60 individually. Bruce Bonnett, 602 Garfield Way, Redlands, CA 92373. Phone (714) 793-9300.

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